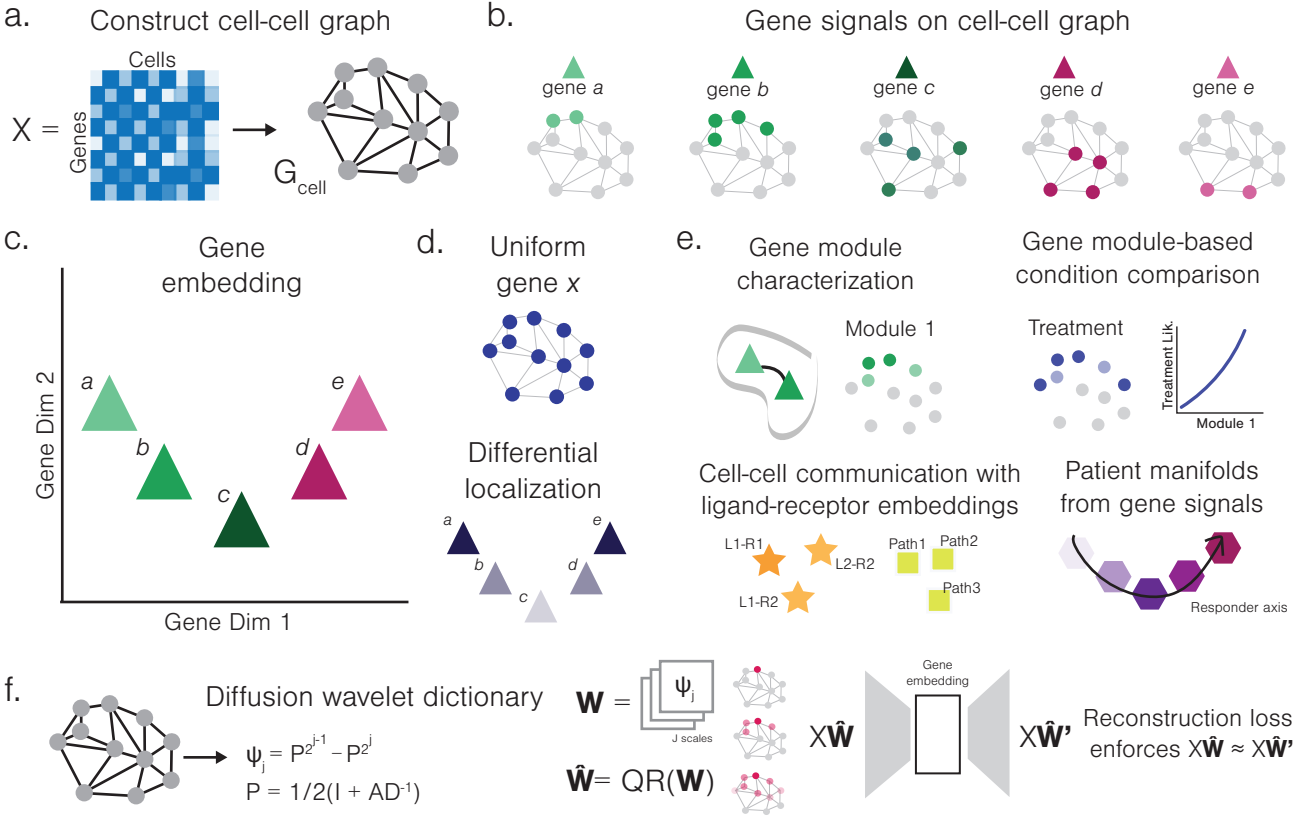
a.
Construct cell-cell graph
Cells
X =
Genes
G cell
b.
Gene signals on cell-cell graph
gene a
gene b
gene c
gene d
gene e
c.
Gene embedding
Gene Dim 2
Gene Dim 1
a
b
c
d
e
d.
Uniform gene x
Differential localization
a
b
c
d
e
e.
Gene module characterization
Module 1
Gene module-based condition comparison
Treatment
Treatment Lik.
Module 1
Cell-cell communication with ligand-receptor embeddings
L1-R1
L2-R2
L1-R2
Path1
Path2
Path3
Patient manifolds from gene signals
Responder axis
f.
Diffusion wavelet dictionary
ψj = P2j-1 − P2j
P = 1/2(I + AD-1)
W =
ψj
J scales
Ŵ= QR(W)
XŴ
Gene embedding
XŴ'
Reconstruction loss enforces XŴ ≈ XŴ'